

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 23-2026: A 28-Year-Old Woman with Nausea, Dizziness, and Metabolic Acidosis

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PRESENTATION OF CASE

Dr. James A. Gordon (Chief Learning Officer): The Peter L. Slavin, MD Academy for Applied Learning in Health Care is delighted to collaborate with the Department of Emergency Medicine in jointly hosting a clinicopathological conference in the Ether Dome, the historic teaching amphitheater of the Massachusetts General Hospital.

Dr. Aaron D'Amore (Emergency Medicine): A 28-year-old woman presented to the emergency department of this hospital because of abdominal pain, dizziness, nausea, and vomiting.

The patient had been in her usual state of generally good health until the night before this presentation, when epigastric pain and vomiting developed abruptly after the consumption of spicy food. Her partner had consumed the same food but did not have similar symptoms. The patient had persistent vomiting and lightheadedness and was unable to maintain adequate hydration; subsequently, she presented to the emergency department of this hospital, accompanied by her partner.

On presentation to the emergency department, the oral temperature was 36.0°C, the blood pressure 122/88 mm Hg, the respiratory rate 20 breaths per minute, the heart rate 88 beats per minute, and the oxygen saturation 100% while the patient was breathing ambient air. She reported ongoing epigastric pain and nausea, and several witnessed episodes of emesis occurred after her arrival at the emergency department, with vomitus that was nonbloody and nonbilious. The patient also reported ongoing dizziness, lightheadedness, and vertigo. A review of systems was negative for fever, chest pain, diarrhea, hematochezia, melena, anorexia, cough, dyspnea, headache, paresthesia, vision symptoms, and motor weakness. Her partner reported that the patient had presented to the emergency department with similar symptoms 2 months earlier; the symptoms had been self-limited and had been treated with antiemetic agents and antacids. The partner described the patient as having a “sensitive stomach” and noted that she often had similar symptoms after eating spicy foods.

Four hours after presentation to the emergency department, the oral temperature was 35.8°C, the blood pressure 139/93 mm Hg, the heart rate 130 beats per minute, the respiratory rate 25 breaths per minute, and the oxygen saturation 99%

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CME



while the patient was breathing ambient air. The weight was 50 kg, and the body-mass index (the weight in kilograms divided by the square of the height in meters) was 18. The patient was upset and agitated. Tachycardia was present without murmur, and the lungs were clear on auscultation. The abdomen was mildly tender on deep palpation of the epigastrium; no masses or hepatosplenomegaly were noted. The neurologic examination showed end-gaze nystagmus on lateral gaze that resolved spontaneously; no other abnormalities were observed on pupillary or extraocular muscle testing. The gait was normal, and the patient had no motor or sensory deficits. Treatment with intravenous lactated Ringer's solution, ondansetron, and famotidine was initiated.

Laboratory test results were notable for a blood bicarbonate level of 10 mmol per liter (reference range, 23 to 32), with normal blood levels of sodium, chloride, and glucose. The anion gap was 21 mmol per liter (reference range, 3 to 17). A venous blood gas analysis was notable for a pH of 7.23 (reference range, 7.30 to 7.40) and a partial pressure of carbon dioxide of 26 mm Hg (reference range, 30 to 50). The levels of alanine aminotransferase, aspartate aminotransferase, bilirubin, alkaline phosphatase, and albumin were normal, as were the results of kidney-function tests. Additional laboratory test results are shown in Table 1. Computed tomography of the abdomen and pelvis, performed after the administration of intravenous iodinated contrast material, showed no abnormalities.

Six hours after presentation to the emergency department, the patient reported increasing dizziness. She appeared to be more anxious and less engaged than she had appeared during the previous interview. Her partner, who was still present, was asked to leave the examination room in order to allow for assessment of the possibility of intimate partner violence contributing to the presentation. After the patient's partner left the room, the patient gradually became more activated and anxious, pacing the room and becoming distraught and tearful, and requested to leave the hospital. She reported no abuse by her partner and that she felt safe and would feel better if he accompanied her while she was in the hospital. The patient was intent on pursuing a self-directed discharge. However, in the context of emotional dysregulation, she did not show sufficient capac-

ity to decide to leave the hospital and was placed under a psychiatric hold and clinical monitoring owing to concern about severe metabolic acidosis. A decision was made to admit the patient to the hospital.

The patient's medical history included constipation and gastroesophageal reflux disease (GERD). She took no prescription medications and reported no use of over-the-counter medications, dietary supplements, or herbal products. She had no known adverse reactions to medications. She lived with a roommate in Massachusetts and reported no use of alcohol, tobacco, or recreational substances. She was born in East Asia and was of East Asian descent; she had moved to the United States a few years before this presentation to attend university. She was in a monogamous relationship with her partner.

Regular laboratory testing performed 6 hours after presentation to the emergency department was notable for a blood bicarbonate level of 9 mmol per liter and a blood urea nitrogen level of 5 mg per deciliter (2 mmol per liter; reference range, 8 to 25 mg per deciliter [3 to 9 mmol per liter]). Blood levels of lactic acid and β -hydroxybutyric acid were within normal limits, and urinalysis results were unremarkable. Blood toxicology studies were negative for ethanol, salicylates, acetaminophen, and tricyclic antidepressant medications, and urine toxicology testing was negative for opioids, amphetamines, cocaine metabolites, and benzodiazepines. The serum osmolality was elevated, at 363 mOsm per kilogram of water (reference range, 280 to 296). Complete laboratory test results are shown in Table 1.

A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Hilarie H. Cranmer: I participated in the care of this patient and am aware of the final diagnosis. This young woman presented to the emergency department with nausea, vomiting, epigastric pain, and dizziness. Laboratory studies revealed a progressively widening anion gap, and a blood gas analysis confirmed the presence of acidemia — findings consistent with high anion-gap metabolic acidosis. Her medical history was unremarkable, and it is notable that she did not use alcohol or have diabetes mellitus. Although the patient was thin, no evidence of starvation or malnutrition

Table 1. Laboratory Data.*

Variable	Reference Range, Adults†	On Presentation, Emergency Department	6 Hr after Presentation, Emergency Department
Hemoglobin (g/dl)	12.0–16.0	14.9	—
Hematocrit (%)	41.0–53.0	45.2	—
White-cell count (per μ l)	4500–11,000	7400	—
Platelet count (per μ l)	150,000–400,000	206,000	—
Sodium (mmol/liter)	135–145	138	138
Potassium (mmol/liter)	3.4–5.0	4.4	4.5
Chloride (mmol/liter)	98–108	107	108
Carbon dioxide (mmol/liter)	23–32	10	9
Anion gap (mmol/liter)	3–17	21	—
Urea nitrogen (mg/dl)	8–25	8	5
Creatinine (mg/dl)	0.60–1.50	0.51	0.40
Calcium (mg/dl)	8.5–10.5	8.5	9.0
Glucose (mg/dl)	70–110	111	103
Aspartate aminotransferase (U/liter)	10–40	27	—
Alanine aminotransferase (U/liter)	10–55	28	—
Alkaline phosphatase (U/liter)	30–100	70	—
Total bilirubin (mg/dl)	0.0–1.2	0.9	—
Direct bilirubin (mg/dl)	0.0–0.3	0.3	—
Albumin (g/dl)	3.3–5.0	5.1	—
Globulin (g/dl)	1.9–4.1	3.2	—
Serum osmolality (mOsm/kg of water)	280–296	—	363
Lactic acid (mmol/liter)	0.5–2.0	—	0.9
β -Hydroxybutyric acid (mmol/liter)	<0.4	—	0.1
Prothrombin time (sec)	10.0–13.0	10.9	—
International normalized ratio	0.9–1.1	1.0	—
Venous blood gases			
pH	7.30–7.40	7.23	7.18
Partial pressure of carbon dioxide (mm Hg)	30–50	26	23
Partial pressure of oxygen (mm Hg)	35–50	77	60

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for bilirubin to micromoles per liter, multiply by 17.1. To convert the values for lactic acid to milligrams per deciliter, divide by 0.1110.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

was noted. She also reported no use of over-the-counter analgesic agents, such as acetaminophen, nonsteroidal antiinflammatory drugs, or aspirin. The patient's similar presentation 2 months earlier had been attributed to GERD and constipation, but no laboratory evaluation had been performed.

This second visit prompted a more thorough workup.

DEFINING THE ACID–BASE DISORDER

The patient's initial laboratory testing revealed a blood bicarbonate level of 10 mmol per liter and

an anion gap of 21 mmol per liter. Given these abnormalities, it is critical to perform a blood gas analysis to determine the patient's acid-base status.¹ The venous blood gas analysis was notable for a pH of 7.23 and a partial pressure of carbon dioxide of 26 mm Hg — findings that were consistent with metabolic acidosis with respiratory compensation. Clinically, this diagnosis was supported by the patient's visible agitation and hyperpnea.²

METABOLIC ACIDOSIS

High anion-gap metabolic acidosis occurs in the presence of excess organic or inorganic acids. In solution, these acids dissociate to release a positively charged proton and a negatively charged conjugate base. The proton binds to bicarbonate, yielding water and carbon dioxide, whereas the resultant conjugate base remains in solution. The conjugate base (the unmeasured anion) is negatively charged, and thus, it replaces the consumed bicarbonate, which results in preservation of electroneutrality without any change in the chloride level and widening of the anion gap. It is notable that this patient's bicarbonate level was lower than expected given the degree of the anion gap, under the clinical approximation that each increase of 1 mmol in the anion gap consumes 1 mmol of bicarbonate. This finding suggests the presence of a superimposed metabolic acidosis with a normal anion gap.

Several mnemonics have been used to aid in developing a differential diagnosis of high anion-gap metabolic acidosis; perhaps the most common of these are “MUDPILES” (methanol, metformin, uremia, diabetic ketoacidosis, paraldehyde, isoniazid, iron, lactic acid, ethylene glycol, and salicylates) and “GOLD MARK” (glycols, oxoproline, L-lactic acid, D-lactic acid, methanol, aspirin, renal failure, and ketoacidosis). The more recent GOLD MARK mnemonic incorporates oxoproline and propylene glycol and omits paraldehyde, which is no longer in use.³ Propylene glycol is a diluent in some common intravenous medications, including lorazepam, diazepam, digoxin, phenobarbital, and phenytoin; it is metabolized in the liver to L-lactic acid and D-lactic acid. The presence of oxoproline (an acetaminophen metabolite) can result in high anion-gap metabolic acidosis with long-term use of acetaminophen, especially in women and in persons who are malnourished.⁴ In addition, cyanide and carbon mon-

oxide poisoning should be considered in the differential diagnosis of high anion-gap metabolic acidosis.

Several diagnoses could easily be discarded on the basis of the MUDPILES and GOLD MARK mnemonics. Uremia was ruled out because the patient had normal kidney function. Lactic acidosis was considered, given the patient's tachycardia and abdominal pain, but the blood level of lactic acid was not elevated. Ketoacidosis due to diabetes, alcohol use, or starvation was eliminated from the differential diagnosis because the patient had a normal blood glucose level, an undetectable blood level of β -hydroxybutyric acid, and an undetectable urine ketone level. The blood levels of ethanol, acetaminophen, and salicylates were undetectable^{5,6}; it is important to consider that these measurements are time-dependent, and previous ingestion cannot be ruled out by laboratory studies alone. However, the normal aspartate aminotransferase and alanine aminotransferase levels and normal international normalized ratio seen in this patient are not consistent with acetaminophen-related hepatotoxicity.

TOXIC ALCOHOL INGESTION

In the context of unexplained anion-gap acidosis, measurement of serum osmolality can provide important information by allowing for calculation of an osmolar gap, which may indicate whether unmeasured osmoles, such as toxic alcohols, are present in the blood. The patient's calculated serum osmolality, based on laboratory values obtained on presentation, was 285 mOsm per kilogram of water (on the basis of the formula of $[2 \times \text{sodium level}] + [\text{glucose level}/18] + [\text{urea nitrogen level}/2.8]$), and the measured serum osmolality was 363 mOsm per kilogram of water, which resulted in an osmolar gap of 78 mOsm per kilogram of water (reference value, <10). A high osmolar gap in the context of high anion-gap metabolic acidosis is highly suggestive of an unmeasured osmotically active substance and, in the context of this patient's clinical presentation, prompted immediate consideration of toxic alcohol ingestion. It is important to note that the osmolar gap tends to be highest early after ingestion and can normalize as the parent compound is metabolized to its toxic organic acid metabolites, which subsequently expands the anion gap — meaning that a normal osmolar gap does not rule out toxicity and that timing of the assessment should be considered.

On the basis of these findings, the most likely cause of this patient's metabolic acidosis was ingestion of a toxic alcohol such as methanol, ethylene glycol, or propylene glycol, with oxoproline (pyroglutamic acid) as an additional consideration. Methanol is metabolized by alcohol dehydrogenase (ADH) to formaldehyde and then to formic acid, which can accumulate and cause metabolic acidosis, vision impairment, optic-nerve damage, and in severe cases, coma and death. Clinical signs may appear 30 minutes to 4 hours after ingestion. Methanol toxicity is a worldwide phenomenon, with an increasing incidence of outbreaks associated with adulterated liquor causing possibly thousands of deaths every year, particularly in low- and middle-income countries where fomepizole therapy is minimally available and ethanol therapy is expensive.⁷⁻¹⁰

Ethylene glycol follows a different pathway: glycolic acid is metabolized in the liver to oxalic acid, which binds to calcium, resulting in calcium oxalate deposition in the kidney. The accumulation of these crystals can lead to proximal tubular necrosis and renal failure. Propylene glycol, unlike the other alcohols, does not produce a toxic metabolite per se but overwhelms normal elimination pathways and can produce high anion-gap metabolic acidosis with an elevated osmolar gap. Oxoproline accumulates when glutathione is depleted — as occurs with long-term acetaminophen use in the context of malnutrition or renal disease; these changes can result in inhibition of the enzyme 5-oxoprolinease and the development of high anion-gap metabolic acidosis, which may be underrecognized. Because this patient did not have any exposure to intravenous medications or large amounts of acetaminophen, we focused our treatment strategy on the management of either methanol or ethylene glycol toxicity.

TREATMENT AMID UNCERTAINTY

Treatment for ingestion of toxic alcohols should not be delayed until a definitive diagnosis is confirmed. Fomepizole, an inhibitor of ADH, should be administered to block the conversion of methanol or ethylene glycol to their toxic metabolites. Because folate is required for the elimination of formic acid in persons with methanol toxicity, and because thiamine and pyridoxine support alternative metabolic pathways in persons with ethylene glycol toxicity, coadministration of B vitamins, including thiamine and leucovorin (also known as

folinic acid), is a low-risk intervention.⁸⁻¹⁰ For this patient, the nephrology service was also consulted for consideration of hemodialysis.

CLINICAL IMPRESSION AND DIAGNOSIS

Dr. Kendra E. Wulczyn: The inpatient nephrology service was consulted to evaluate the patient's metabolic acidosis. Initial laboratory test results showed combined anion-gap and non-anion-gap metabolic acidosis with appropriate respiratory compensation. The blood level of creatinine was normal, given the patient's age and body habitus. The blood level of lactic acid was normal, and both the blood level of salicylates and the urine level of ketones were undetectable.

In the case of anion-gap metabolic acidosis of uncertain cause, serum osmolality should be measured promptly to assess for possible toxic alcohol ingestion. This patient's osmolar gap was nearly eight times the upper limit of the normal range, and the blood ethanol level was undetectable. Given the magnitude of the osmolar gap, along with the presence of an elevated anion gap, ingestion of a toxic alcohol was considered to be the most likely diagnosis in this patient.

Methanol and ethylene glycol are metabolized to strong acids, which initially creates an osmolar gap from the parent alcohol and later, progressively worsening high anion-gap metabolic acidosis with intermediate combined osmolar gap and anion-gap metabolic acidosis. Whether this patient's reported dizziness reflected a degree of vision impairment and was indicative of methanol toxicity could not be reliably assessed, given her level of agitation while she was in the emergency department. The patient reported that she had not ingested any specific substance, and the patient-reported history was not useful in distinguishing between the two toxic alcohols.

Empirical treatment with fomepizole was initiated, and a continuous infusion of sodium bicarbonate was started to correct acidemia. Within a few hours after a sample of the patient's blood had been obtained for methanol testing, the blood level of methanol became available and was found to be 190 mg per deciliter (reference value, negative), which confirmed the diagnosis of methanol toxicity and led to initiation of emergency hemodialysis. Owing to the patient's level of agitation and confusion, she was unable to consent to

placement of a temporary hemodialysis catheter and initiation of hemodialysis; however, given the severity of the intoxication and the need for prompt treatment, the medical team chose to proceed.

DR. HILARIE H. CRANMER'S DIAGNOSIS

Toxic alcohol ingestion.

CLINICAL DIAGNOSIS

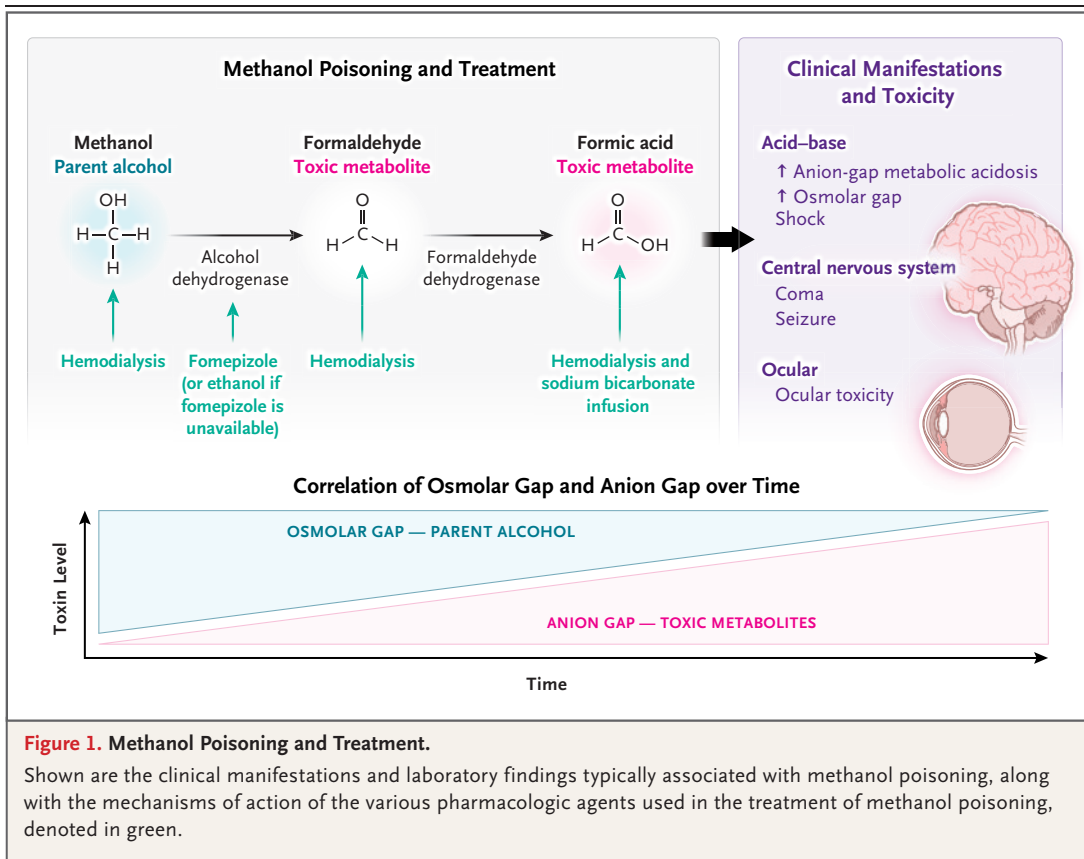
Methanol toxicity.

DISCUSSION OF TOXICOLOGIC MANAGEMENT

Dr. Timothy B. Erickson: Methanol is found in many household items, such as automotive supplies and cleaning solvents. When ingested, methanol is rapidly absorbed in the gastrointestinal tract; consequently, gastrointestinal decontamination with oral activated charcoal has minimal value.

Gastric lavage is not indicated, but gastric suctioning through a nasogastric or orogastric tube may be considered in cases of intentional large-volume overdose immediately after ingestion. In addition to standard stabilization and resuscitation measures, management of methanol toxicity involves inhibiting the production of formic acid with the administration of the ADH inhibitor fomepizole (or 4-methylpyrazole), correcting the acidosis with the administration of sodium bicarbonate, and eliminating methanol and its toxic metabolites with hemodialysis (Fig. 1). Therapy should be initiated early in the clinical course on the basis of strong diagnostic suspicion and should not be delayed while awaiting measurement of blood levels of methanol by the laboratory.

The strongest predictor of complications and death in patients who have ingested methanol is the degree of acidosis, with high mortality observed at a pH of less than 7.00. A comatose state at presentation also predicts high mortality and long-term neurologic consequences.¹¹ Because the degree of acidosis directly correlates with severity and outcome, treatment with intravenous sodium



bicarbonate is recommended when the pH is less than 7.30 and can be continued until the pH normalizes. Large amounts of sodium bicarbonate may be necessary for the correction of acidosis owing to the metabolism of the parent alcohol into its toxic acid.

Further production of formic acid is prevented by inhibiting ADH with the administration of fomepizole. Specific indications for initiating ADH blockade include a blood methanol level of greater than 20 mg per deciliter, a history of methanol poisoning with an osmolar gap of greater than 10 mOsm per kilogram of water, suspected methanol ingestion with an arterial pH of less than 7.30, or a blood bicarbonate level of less than 20 mmol per liter. Of note, adjustment of the frequency of fomepizole administration is necessary during hemodialysis.¹² Delays in ADH blockade can lead to worsening acidosis, vision disturbances, and permanent neurologic consequences. In addition, confirmatory testing for methanol levels should be obtained from a reference laboratory because many hospitals lack the resources to test for toxic alcohols, which can lead to a delay in confirming the diagnosis.¹³ When assessing for methanol toxicity, it is prudent to also obtain confirmatory testing for possible concomitant ethylene glycol toxicity.

If fomepizole is not available, ethanol therapy can be used to inhibit ADH by maintaining the blood ethanol level between 100 and 150 mg per deciliter. As a competitive inhibitor, ADH has an affinity for ethanol that is 10 times as high as its affinity for methanol.¹⁴ However, determining the dose of ethanol to be administered is challenging and can cause worsening central nervous system depression, hypotension, vomiting, and hypoglycemia — particularly in children or malnourished patients.¹⁴

In patients with severe toxicity, hemodialysis is the definitive therapy for elimination of methanol and its toxic metabolites. Timely hemodialysis removes the parent alcohol and its metabolites, corrects acidosis, and aids in fluid and electrolyte management.¹⁵ Intermittent hemodialysis is preferred over continuous renal replacement therapy, but the latter is acceptable if hemodialysis is not available.

Hemodialysis is indicated for any of the following conditions that are attributed to methanol poisoning: coma, seizures, new vision deficits, metabolic acidosis with an arterial pH of 7.15 or

lower, persistent metabolic acidosis despite adequate supportive measures and antidotes, a serum anion gap higher than 24 mmol per liter, or a methanol level greater than 50 mg per deciliter in the absence of an ADH blocker.¹⁶ If the methanol level is unavailable, an elevated osmolar gap may also be informative when considering hemodialysis. Recommended criteria for discontinuing hemodialysis or ADH inhibition are a normal acid–base status and a methanol level of less than 20 mg per deciliter. Vision abnormalities are not an indication for continued hemodialysis after correction of the acid–base disturbance and removal of methanol because there is no specific treatment for irreversible methanol-induced optic-nerve damage. Small trials have investigated the use of erythropoietin and glucocorticoids for methanol-induced optic neuropathy, with mixed results¹⁷; thus, routine use is not currently recommended. Lastly, leucovorin (folinic acid) may aid in the elimination of formic acid and can be administered until the methanol has been eliminated and acidosis resolves; however, data are lacking from recent trials in humans to support its clinical efficacy.

DISCUSSION OF PSYCHIATRIC MANAGEMENT

Dr. Scott R. Beach: Broadly speaking, toxic substances can enter the body in one of three ways: intentional ingestion by the patient (with the goal of intoxication, suicide, or nonsuicidal self-injury), intentional administration by a third party (with a possible goal of homicide or induction of illness or as a practical joke), and accidental ingestion (by means of contamination [e.g., poison accidentally mixed with food or drink], substitution [e.g., drinking from the wrong container], or improper preparation [e.g., fermentation leading to an unanticipated by-product]) (Table 2). Many of these possibilities have their own differential diagnoses for underlying psychiatric illness. For example, suicide could be motivated by underlying bipolar disorder, major depression, or a personality disorder, whereas illness induction could occur with a personality disorder or factitious disorder. In this patient's case, consideration of intentional ingestion and intentional administration prompted us, as the psychiatric consultants, to “think dirty” — that is, to consider the presence of possible hidden motives that could have driven the patient

Table 2. Psychiatric Considerations for Identifying the Cause of Toxic Substance Intoxication.

Intentional ingestion by the patient: think about the “why”
Suicide attempt
Nonsuicidal self-injury
Intoxication
Intentional administration by a third party: think about the “why”
Homicide
Induction of illness
Prank
Accidental ingestion: think about the “how”
Contamination (e.g., poison accidentally mixed with food or drink)
Substitution (e.g., drinking from the wrong container)
Improper preparation (e.g., fermentation leading to an unanticipated by-product)

or others to withhold information from (or deceive) providers.¹⁸

When intentional ingestion with suicidal intent was considered in this patient, a psychiatric hold was initially maintained; we reasoned that the risk of the patient leaving the hospital after a possible suicide attempt outweighed the risk of keeping her against her will, especially given the severity of her condition. We interviewed her alone, using nonjudgmental language and making gentle assumptions¹⁹ when asking about possible ingestion. In addition, collateral information was obtained, and we assessed for concurrent psychiatric conditions. The patient was initially evasive and reacted oddly to questioning, probably owing to an intoxicated state. Eventually, given that she repeatedly indicated during a 48-hour period that no intentional ingestion had occurred, and given that there was insufficient evidence to support intentional ingestion, the legal hold was removed.

The possibility of intentional administration by a third party prompted an important safety question that psychiatrists consider infrequently: Is someone else an imminent danger to this patient? Access and motive were considered, and we interviewed both the patient and her social supports separately, closely examining elements of the narratives that did not align. We also considered broader contextual possibilities, including human trafficking, but considered this to be an unlikely possibility. In collaboration with le-

gal counsel at the hospital, we opted to warn the patient about this possibility but not to pursue further action.

Finally, we considered possible mechanisms for accidental ingestion. The search focused on foods and beverages consumed in the previous 48 hours, but the patient and her partner reported no consumption of anything fermented or alcohol-based and consistently reported no exposure to solvents, windshield-wiper fluid, antifreeze, cleaning products, or other household sources of methanol. Because the patient was of Asian descent and therefore could have had reduced levels of ADH, we considered the possibility that this circumstance could theoretically have led to higher-than-expected levels of methanol in the context of a small ingestion, although this consideration was speculative.²⁰

HOSPITAL COURSE

Dr. Wulczyn: Once the methanol level was found to be markedly elevated, the decision was made to initiate emergency hemodialysis. Given the patient's ongoing anxiety and agitation, she was transferred to the medical intensive care unit, where a continuous infusion of dexmedetomidine was initiated for placement of a temporary hemodialysis catheter. On the basis of the patient's initial methanol level and her weight, we estimated that 6 hours of hemodialysis would be sufficient to clear methanol and its toxic metabolites.

Because methanol levels can take hours to obtain, we used surrogate markers of methanol to approximate when it had been fully cleared from the patient's circulation. We stopped hemodialysis once the blood pH and blood bicarbonate level had normalized and once the osmolar gap fell below 10. When the serial methanol levels eventually became available, they confirmed a rapid decline, from 190 mg per deciliter to below the limit of detection within approximately 5 hours after initiation of hemodialysis.

The source of the methanol in this case was never definitively established. The patient consistently reported no ingestion of any specific substance and expressed that she did not have concern about any risk of harm from others. The psychiatric hold, which had been in place since her presentation given the inability to rule out

intentional ingestion, was lifted on hospital day 3, and the patient pursued a self-directed discharge shortly thereafter.

FINAL DIAGNOSIS

Methanol toxicity.

This case was presented at Emergency Medicine Grand Rounds in the Ether Dome at Massachusetts General Hospital in collaboration with the Peter L. Slavin, MD Academy for Applied Learning in Health Care, founded with generous support from the Lunder

Foundation of Portland, Maine, to improve patient care through excellence in education.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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